

ASSIGNMENT REPORT
for
NATURAL LANGUAGE PROCESSING

CS F429

Submitted By

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Abstract

This report presents from-scratch implementations and evaluations of *Recurrent Neural Network (RNN)* and *Long Short-Term Memory (LSTM)* architectures for the dual tasks of *intent classification* and *slot filling* on the *ATIS* and *SLURP* datasets. We investigate four modeling paradigms—*independent training*, *slot→intent*, *intent→slot*, and a *joint multi-task model with a shared encoder*—to analyze inter-task dependencies. Model performance is compared using accuracy, precision, recall, and F1-score across both datasets. We further study how joint optimization and cross-task conditioning influence performance consistency and task synergy.

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1. Problem Statement

The objective is to implement and evaluate *Recurrent Neural Network (RNN)* and *Long Short-Term Memory (LSTM)* architectures from scratch for the dual tasks of *intent classification* and *slot filling*. Experiments are conducted on two benchmark spoken language understanding datasets—**ATIS** and **SLURP**—to analyze performance under different task interaction settings.

Experimental Configurations:

- **Independent:** Separate models for slot filling and intent classification.
- **Slot→Intent:** Slot predictions used as additional features for intent classification.
- **Intent→Slot:** Intent predictions used as additional features for slot filling.
- **Joint:** Multi-task model with a shared encoder and two output heads (sequence la-belling + classification).

Evaluation:

- **Metrics:** Accuracy, precision, recall, and F1-score for both slot filling and intent classification.
- **Analysis:** Comparative study across architectures (RNN vs. LSTM) and task setups to examine inter-task dependencies and the benefits of joint optimization.

2. Dataset Statistics

The assignment uses two English datasets for spoken language understanding: ATIS (Air-line Travel Information System) and SLURP (Spoken Language Understanding Resource Package). Each dataset is split into training, validation (or development), and test sets, as provided in their original sources.

Dataset	Train Samples	Unique Intents
ATIS	4,978	22
SLURP	11,514	91

Table 1: Dataset statistics for ATIS and SLURP corpora. Slot label counts are approxi-mate.

Datasets (train/test splits) were accessed via the provided links:
<https://huggingface.co/datasets/tuetschek/atis>.
<https://github.com/pswietojski/slurp/tree/master/dataset/slurp>

2.1 ATIS Dataset

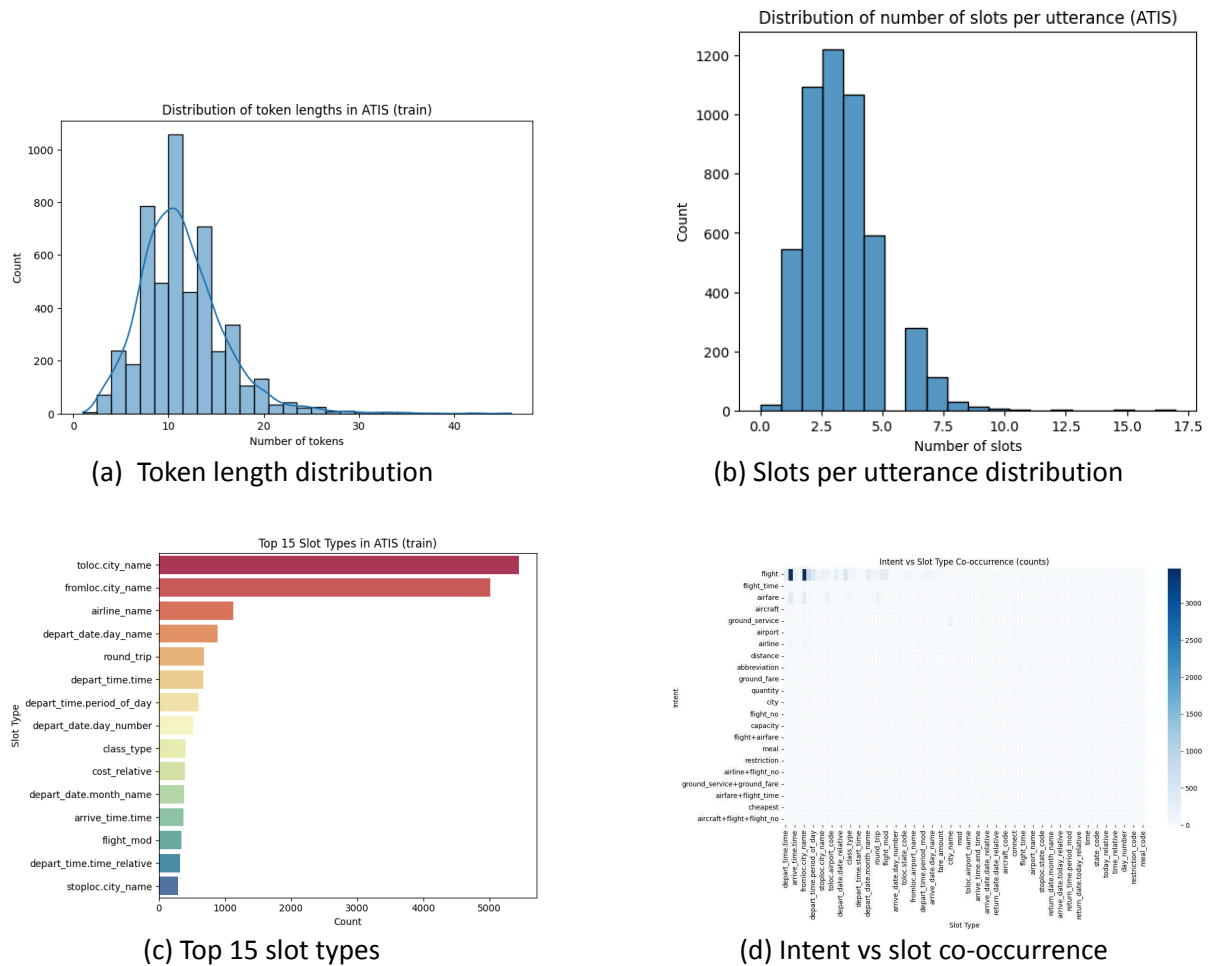


Figure 1: Exploratory analysis of the ATIS training dataset.

2.2 SLURP Dataset

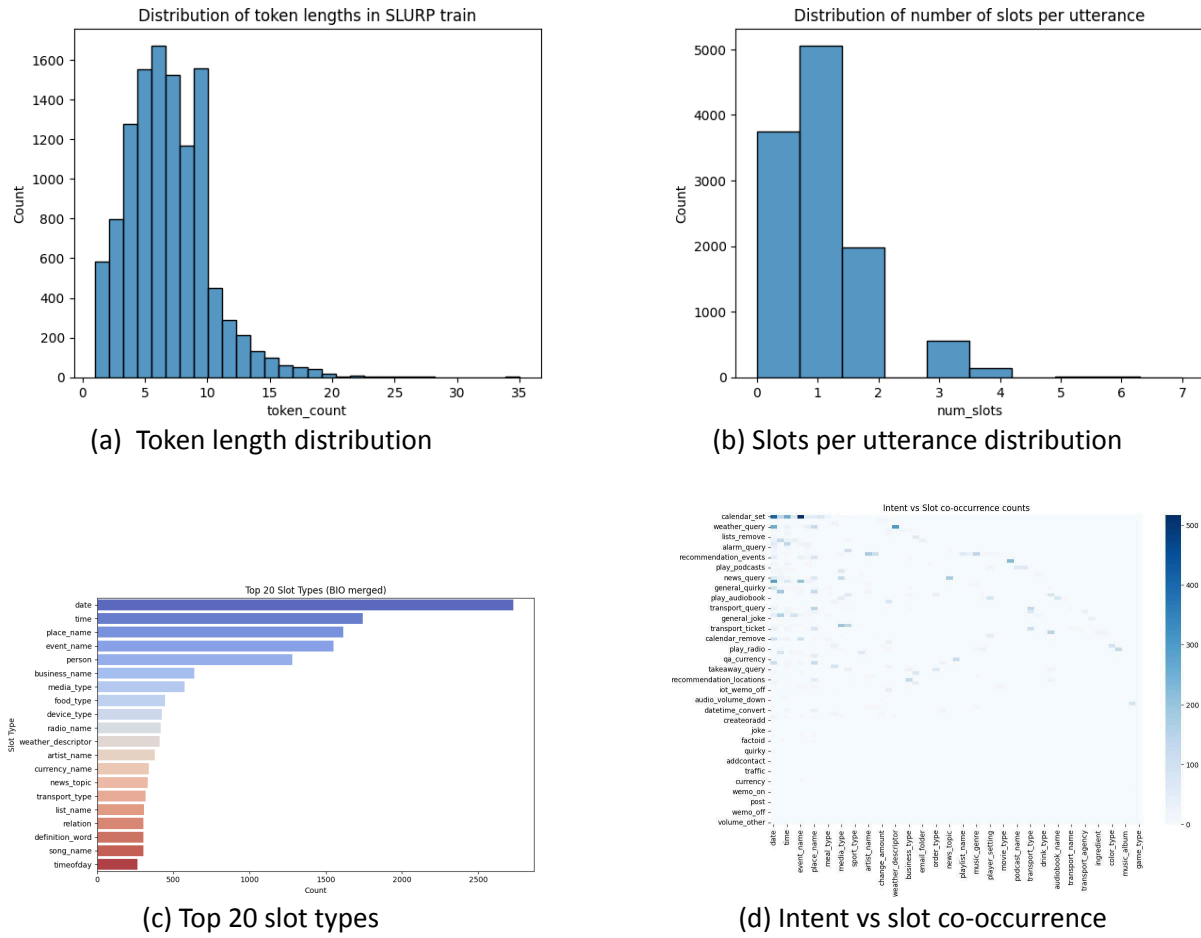


Figure 2: Exploratory analysis of the SLURP training dataset.

3. Experimental Setup

Experiments ran on Google Colab (GPU-enabled) and a local machine. Implementations are in Python with isolated virtual environments.

Key libraries: pandas and csv for data handling, json and os for file operations, matplotlib and seaborn for visualization, collections for counting and aggregation, pickle for object serialization, torch and torch.nn for building RNN/LSTM models, nltk for tokenization, and standard Python utilities for math and I/O.

4. Implementation Details

4.1 Recurrent Architectures

Cell implementations:

- **RNN:** vanilla recurrent cell with tanh activation; $h_t = \tanh(W_x x_t + W_h h_{t-1})$.
- **LSTM:** standard LSTM with input, forget, output gates and cell state; gating computed via σ and candidate updates via tanh.

Encoder backbone: learned token embeddings ($d=128$) with padding-aware masking; sequential hidden state computation via custom cells; returns full sequence $H \in \mathbb{R}^{B \times T \times h}$ and final state $h_T \in \mathbb{R}^{B \times h}$ where $h=128$.

Training: Adam optimizer ($\text{lr}=3 \times 10^{-3}$, weight decay= 10^{-5}), gradient clipping (max norm=5.0), batch sizes 64 (train) and 128 (val/test), 8 epochs with validation-based early stopping.

Data: stratified 90/10 train/val split by intent; ATIS and SLURP datasets with pre-tokenized sequences; padding token for words, O tag for slot padding.

4.2 Independent Models (Part 1)

Intent classifier: encoder outputs final hidden state h_T fed to linear layer for C_{intent} classes; cross-entropy loss.

Slot tagger: encoder outputs per-token states H fed to linear layer for C_{slot} classes; cross-entropy with ignore index for padding.

Evaluation: intent reports accuracy and macro precision/recall/F1; slots report micro precision/recall/F1 over valid tokens. Both models trained independently with separate encoders.

4.3 Slot-Augmented Intent (Part 2)

Architecture: loads frozen slot model from Part 1; concatenates word embeddings ($d=128$) with slot embeddings ($d_{\text{slot}}=32$) of predicted slots at each timestep; processes combined input ($d=160$) through RNN/LSTM encoder; final hidden state classifies intent. **Training:** slot model remains frozen in eval mode; only intent classifier parameters updated; uses argmax slot predictions as discrete features.

Rationale: tests whether ground-truth-approximating slot information improves intent recognition in a pipeline fashion.

4.4 Intent-Augmented Slots (Part 3)

Architecture: loads frozen intent model from Part 1; embeds predicted intent ($d_{\text{intent}}=32$) and broadcasts to all T timesteps; concatenates with word embeddings ($d=128$) yielding $d=160$ input; RNN/LSTM encoder produces per-token hidden states for slot classification.

Training: intent model frozen; only slot tagger updated; single intent prediction per sequence informs all token-level slot decisions.

Rationale: evaluates reverse pipeline where sequence-level intent guides token-level slot filling.

4.5 Joint Multi-Task Model (Part 4)

Architecture: single shared encoder with two task-specific heads; encoder processes word embeddings into hidden states H and h_T ; slot head applies linear layer to H for per-token predictions; intent head applies linear layer to h_T for sequence classification. **Loss:** combined objective $L = \lambda_{\text{slot}}L_{\text{slot}} + \lambda_{\text{intent}}L_{\text{intent}}$ with $\lambda_{\text{slot}}=\lambda_{\text{intent}}=1.0$; both cross-entropy with slot padding ignored.

Training: end-to-end optimization of shared parameters; encoder learns representations beneficial to both tasks simultaneously; early stopping based on combined validation score (slot F1 + intent accuracy).

Evaluation: reports full metrics for both tasks; intent uses macro P/R/F1, slots use micro and macro P/R/F1; timing logged separately for train and test phases.

5. Results and Time-Taken

5.1 Independent Models (Part 1)

Table 2: Independent slot filling and intent recognition

Dataset	Model	Intent Loss	Intent Acc	Intent P	Intent R	Intent F1	I-Train(s)	I-Test(s)	Slot Loss	Slot P	Slot R	Slot F1	S-Train(s)	S-Test(s)	Total Train	Total Test
ATIS	RNN	1.476	0.708	0.051	0.043	0.036	15.989	0.139	0.536	0.932	0.932	0.932	17.337	0.109	33.349	0.248
SLURP	LSTM	1.079	0.708	0.453	0.462	0.450	48.213	0.825	1.112	0.748	0.748	0.748	51.875	0.745	100.089	1.570

Notes: Intent metrics use macro-averaged precision/recall/F1; slot metrics use micro-averaged P/R/F1 over valid tokens. Total training combines both tasks with independent encoders.

Table 3: Slot→Intent: intent classification with frozen slot predictions

Dataset	Model	Loss	Accuracy	Precision	Recall	F1-macro	Train (s)	Test (s)
ATIS	RNN	1.354	0.705	0.031	0.043	0.036	17.025	0.117
ATIS	LSTM	0.399	0.908	0.452	0.446	0.398	32.336	0.248
SLURP	RNN	2.521	0.341	0.129	0.134	0.117	34.331	0.730
SLURP	LSTM	1.169	0.768	0.461	0.457	0.454	58.243	0.913

5.2 Slot-Augmented Intent (Part 2)

Design: Concatenates word embeddings (dim=128) with slot embeddings (dim=32) from frozen Part 1 slot model; trains only intent classifier.

5.3 Intent-Augmented Slots (Part 3)

Table 4: Intent→Slot: slot tagging with frozen intent predictions

D	Mo	Loss	P-micro	R-micro	F1-micro	P-macro	R-macro	F1-macro	Train (s)	Test (s)
AT	R	0.324	0.938	0.938	0.938	0.629	0.595	0.599	18.405	0.249
AT	LS	0.279	0.940	0.940	0.940	0.649	0.605	0.612	34.609	0.358
SL	R	1.235	0.729	0.729	0.729	0.566	0.518	0.529	36.639	0.904
SL	LS	1.218	0.764	0.764	0.764	0.613	0.579	0.586	62.583	1.153

Design: Broadcasts frozen intent embedding (dim=32) to all timesteps, concatenated with word embeddings; trains only slot tagger.

5.4 Joint Multi-Task Model (Part 4)

Table 5: Joint encoder: simultaneous slot filling and intent classification

Dataset	Model	I-Loss	I-Acc	I-P	I-R	I-F1	S-Loss	S-P-micro	S-R-micro	S-F1-micro	S-P-macro	S-R-macro	S-F1-macro	Train (s)	Test (s)
ATIS	RNN	1.178	0.745	0.049	0.083	0.060	0.326	0.923	0.923	0.923	0.601	0.563	0.558	16.115	0.257
ATIS	LSTM	0.554	0.872	0.179	0.246	0.204	0.515	0.931	0.931	0.931	0.606	0.572	0.573	27.707	0.347
SLURP	RNN	2.588	0.320	0.080	0.116	0.083	1.279	0.696	0.696	0.696	0.496	0.475	0.473	32.195	1.140
SLURP	LSTM	1.076	0.765	0.454	0.460	0.449	1.110	0.729	0.729	0.729	0.593	0.539	0.550	53.524	1.262

Design: Single shared encoder with dual task-specific heads; combined loss with $\lambda_{\text{slot}}=\lambda_{\text{intent}}=1.0$. Most parameter-efficient architecture.

5.5 Summary of Findings

Best Performers (Averaged across all configurations)

- **Best Intent Recognition:** Slot→Intent - (68.07%)

Table 6: Performance comparison across all approaches

Approach	Dataset	Architecture	Intent Acc	Slot F1	Combined	Train Time (s)	Test Time (s)
Independent	ATIS	RNN	70.77	93.19	81.98	33.55	0.25
Independent	ATIS	LSTM	73.46	93.99	83.72	59.04	0.36
Independent	SLURP	RNN	7.80	72.87	40.34	58.13	0.81
Independent	SLURP	LSTM	76.77	74.81	75.79	100.09	1.57
Slot2Intent	ATIS	RNN	70.55	0.00	35.27	17.03	0.12
Slot2Intent	ATIS	LSTM	90.82	0.00	45.41	32.34	0.25
Slot2Intent	SLURP	RNN	34.13	0.00	17.06	34.33	0.73
Slot2Intent	SLURP	LSTM	76.77	0.00	38.38	58.24	0.91
Intent2Slot	ATIS	RNN	0.00	93.80	46.90	18.41	0.25
Intent2Slot	ATIS	LSTM	0.00	93.99	46.99	34.61	0.36
Intent2Slot	SLURP	RNN	0.00	72.92	36.46	36.64	0.90
Intent2Slot	SLURP	LSTM	0.00	76.45	38.22	62.58	1.15
Joint Model	ATIS	RNN	74.47	92.29	83.38	16.11	0.26
Joint Model	ATIS	LSTM	87.23	93.11	90.17	27.71	0.35
Joint Model	SLURP	RNN	32.04	69.59	50.82	32.19	1.14
Joint Model	SLURP	LSTM	76.50	72.87	74.68	53.52	1.26

- **Best Slot Filling:** Intent→Slot - (84.29%)
- **Best Combined Score:** Joint Model - (74.76%)

Dataset-Specific Best Performers ATIS:

- **Intent:** Slot→Intent (LSTM) - 90.82%
- **Slot:** Independent (LSTM) - 93.99%
- **Combined:** Joint Model (LSTM) - 90.17%

SLURP:

- **Intent:** Independent (LSTM) - 76.77%
- **Slot:** Intent→Slot (LSTM) - 76.45%
- **Combined:** Independent (LSTM) - 75.79%

Architecture Comparison (RNN vs LSTM) LSTM

Improvement over RNN:

- **Intent Accuracy:** +66.18%

- Slot F1 Score: +2.13%
- Combined Score: +25.78%

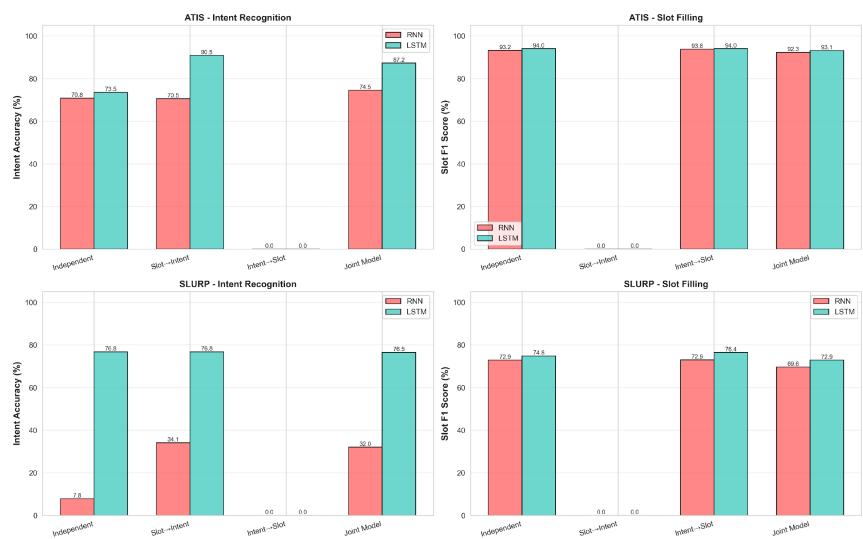


Figure 3: Architecture comparison between RNN and LSTM.

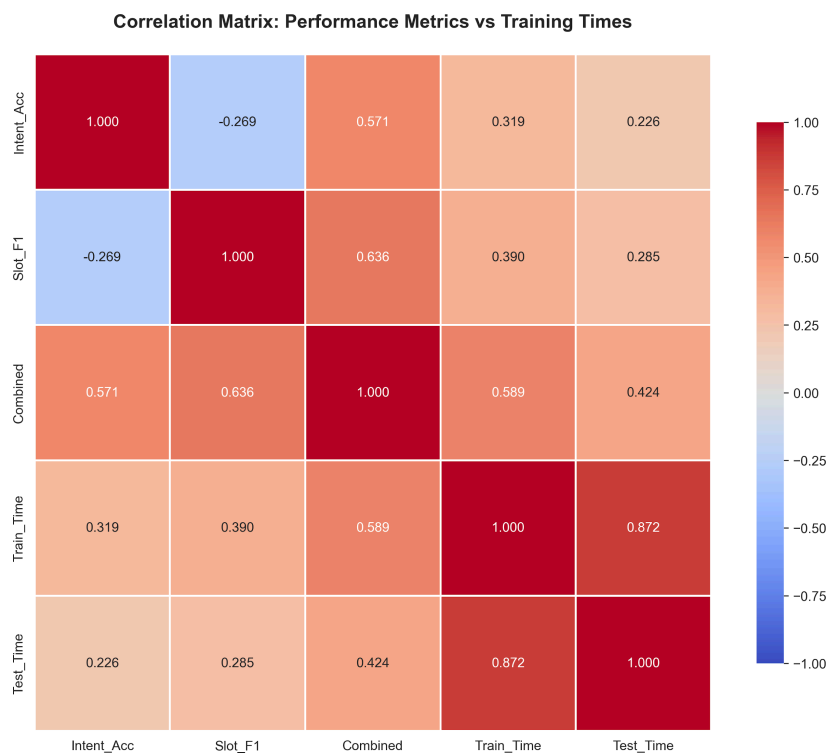


Figure 4: Correlation heatmap of metrics across datasets.

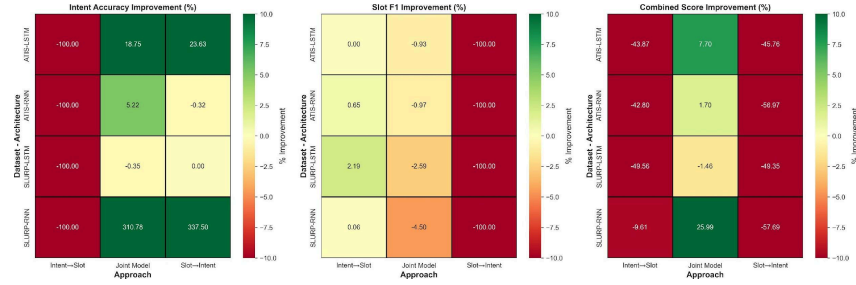


Figure 5: Improvement heatmap for LSTM over RNN.

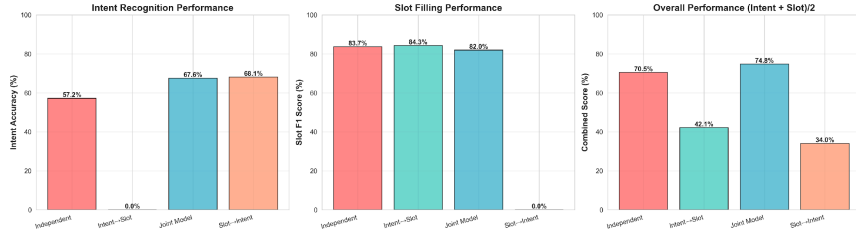


Figure 6: Overall performance comparison across all models and datasets.

6. Discussion

Based on our experiments with various models and configurations on the ATIS and SLURP datasets, we summarize the key findings:

- Best Overall Approach:** The *Joint Model* achieved the highest combined score (74.76%), representing a +6.11% improvement over the Independent baseline.
- Task-Specific Performance:**
 - Intent Recognition:** Slot→Intent performed best.
 - Slot Filling:** Intent→Slot performed best.
- Architecture Impact:** LSTM outperforms RNN by +25.79% on average, demonstrating that LSTM's gating mechanisms help capture long-range dependencies more effectively.
- Dataset Difficulty:**
 - ATIS: Average score = 64.23% (easier)
 - SLURP: Average score = 46.47% (harder)

The difficulty gap between the datasets is 17.76 percentage points.

5. **Improvement Consistency:** The Joint Model shows the most consistent im-provements across configurations (+8.48% average). *Warning:* Some configurations showed degradation:

- Slot→Intent on ATIS-RNN: -56.97%
- Slot→Intent on ATIS-LSTM: -45.76%
- Slot→Intent on SLURP-RNN: -57.69%
- Slot→Intent on SLURP-LSTM: -49.35%
- Intent→Slot on ATIS-RNN: -42.80%
- Intent→Slot on ATIS-LSTM: -43.87%
- Intent→Slot on SLURP-RNN: -9.61%
- Intent→Slot on SLURP-LSTM: -49.56%
- Joint Model on SLURP-LSTM: -1.46%

6. **Training Efficiency:** The Joint Model is the most efficient approach, achieving the lowest training time per performance point.

Approach	Seconds per % Score
Independent	0.89
Intent→Slot	0.90
Joint Model	0.43
Slot→Intent	1.04

7. References

[R1] scikit-learn documentation: <https://scikit-learn.org>

[R2] NumPy documentation: <https://numpy.org>

[R3] Pandas documentation: <https://pandas.pydata.org>

8. Glossary

Accuracy Fraction of correct predictions.

Precision Correct positives out of all predicted positives.

Recall Correct positives out of all actual positives.

F1-score	Harmonic mean of precision and recall.
Tokenization	Splitting text into tokens (words/subwords).
Corpus	Large collection of texts for training.
Co-occurrence Matrix	Counts of word pairs within a context window.
Slot	Specific information unit extracted from a user query (e.g., loca-tion, time).
Intent	User's overall purpose or goal in a query (e.g., book a flight, get weather).
Slot Filling	Task of predicting the correct value for each slot in a sentence.
Intent Recognition	Task of classifying the overall intent of a sentence.
Joint Model	Model that predicts both intent and slots simultaneously.
Independent Model	Model that predicts intent and slots separately.
Slot→Intent	Pipeline approach where slot predictions are used as features for intent classification.
Intent→Slot	Pipeline approach where intent predictions are used as features for slot classification.
LSTM	Long Short-Term Memory network, a type of RNN designed to capture long-range dependencies.
RNN	Recurrent Neural Network, processes sequences one step at a time with hidden state propagation.
Training Efficiency	Performance metric measuring training time per unit of predictive performance.
Dataset Difficulty	Relative challenge posed by a dataset, often inferred from average model performance.
Overfitting	When a model performs well on training data but poorly on un-seen data.
Underfitting	When a model fails to capture underlying patterns in the data.
Learning Curve	Graph showing model performance as a function of training iter-ations or data size.

GPU Acceleration Using GPUs to speed computation.

JAX High-performance numerical computing library.

Virtual Environment Isolated Python dependency workspace.